02-620 Week 9

Machine Learning for Scientists

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Mixture Models

Given samples $x^1, \cdots, x^N$, where each $x^i = [x_{i1}, \cdots, x_{iJ}]$ for N samples and J variables, we want to find K clusters.

- Our data matrix here has N samples (rows), and J variables (columns).
- We are clustering rows together.

In the context of biology, we might be

- clustering genotype data collected for N individuals, J loci.
- clustering expression data of different genes at differnet times.
- etc.

Difference between K-means and Mixture Models

K-means gives hard assignments, Mixture Models give soft assignments.

- When K-means assigns a sample to a given cluster, the sample receives an integer label of the cluster. The issue is that hard clustering does not reflect uncertainty of cluster assignment.
- When a mixture model assigns a sample to a given cluster, it gives a probability for all the different clusters. The actual cluster assignment is the largest probability.
  - Since these are probabilities, the sum of the assignments for one sample must be 1.

A mixture model is a generalization of K-means.

Gaussian Mixture Models

Gaussian Mixture Models use multivariate Gaussians to model data distribution within a cluster.

- A cluster is defined by $N(\mu, \Sigma)$, where in a D-dimensional example, $\mu \in \mathbb{R}^{D \times 1}$ represents the mean of a cluster, and $\Sigma \in \mathbb{R}^{D \times D}$ represents variance.

Gaussian Mixture models data distribution with multiple clusters.

Each data point has:

- (Latent) cluster label: $c \in \{1, \cdots, K\}$, and
- (Observed) data value: $x \in \mathbb{R}^J$

S1: Cluster label, $p(c = i) = \pi_i$ (for example with 3 clusters)

- $p(c = 1) = \pi_1$
- $p(c = 2) = \pi_2$
- $p(c = 3) = \pi_3$

S2: data given label, $p(x|c = i) \sim N(\mu_i, \Sigma_i)$

- $p(x|c = 1) \sim N(\mu_1, \Sigma_1)$
- $p(x|c = 2) \sim N(\mu_2, \Sigma_2)$
- $p(x|c = 3) \sim N(\mu_3, \Sigma_3)$

Overall distribution:

$$p(x) = \sum_{i=1}^{3} p(x, c = i) = \sum_{i=1}^{3} p(x|c = i)p(c = i)$$

Mixture of Gaussians

The probability model for mixture of K Gaussians is:

$$p(x) = \sum_{i=1}^{K} p(x, c = i) = \sum_{i=1}^{K} p(x|c = i)p(c = i)$$

- Mixing proportions $p(c = i) = \pi_i$, for $i = 1, \ldots, K$ clusters
- Mixture component parameters μ_i, Σ_i: mean and covariance of Gaussians distributions for $i = 1, \ldots, K$ clusters

Mixture Model as a Bayesian Network

- A mixture model is a special case of a Bayesian network
  - c: cluster labels
  - x: observed data
- What is the joint probability distribution?

$$\begin{aligned} \rightarrow P(x, c) &= P(c) \times P(x|c) \\ \rightarrow P(x) &= \sum_{i=1}^{k} P(x|c = i)P(c = i) \\ &= \sum_{i=1}^{k} P(x, c = i) \end{aligned}$$

- A Hidden Markov Model is many mixture models threaded over time.

Model Summary

$x^1, \cdots, x^N$ generated IID as:

- (Latent variable) cluster label: $c^n \sim \text{Categorical}(\pi_1, \cdots, \pi_k)$
- (Observed) $[x^n|c^n = i] \sim N(\mu_i, \Sigma_i)$
- Parameters: π_i, μ_i, Σ_i for each component i
- Distribution: $p(x^n) = \sum_{i=1}^{K} p(c^n = i)p(x^n|c^n = i)$

Mixture Model Inference

Given observed data x, can you infer cluster labels c?

- Assume you have a mixture model estimated from data.
- How can you infer the cluster labels for each sample?

The model is

$$p(x, c) = p(x|c)p(c)$$

- For sample x, infer cluster label c
- More specifically, infer $p(c = i|x)$

Using bayes rule,

$$p(c = i|x) = \frac{p(x|c = i)p(c = i)}{p(x)} = \frac{p(x|c = i)p(c = i)}{\sum_{i=1}^{K} p(x|c = i)p(c = i)}$$

Recall

$$p(c = i) = \pi_i \quad p(x|c = i) \sim N(\mu_i, \Sigma_i)$$

Inference Summary

- Model: π_i, μ_i, Σ_i for each component i
- Question: given x^n, what's i with the largest $p(c^n = i|x^n)$
- Procedure:
  - For each i, compute $p(x^n|c^n = i)p(c^n = i)$
  - Select i with the largest such value

Mixture Model Learning

Learning parameters:

- If we have fully observed data, (c, x), we can use MLE!
- If we have partially observed data x, no observations for c, then we use the EM algorithm!

Model selection:

- How many clusters do we want?

Learning with Fully Observed Data

We have observed data (N samples, J variables), and observed cluster labels (N samples)

- Given samples $x^1, \cdots, x^N$, where each $x^n = [x_i^n, \cdots, x_J^n]$, for N samples and J variables.
- Assume **cluster labels** $c^1, \cdots, c^N$ are known.
- Then, we can perform MLE

$$\begin{aligned}
&\arg\max_{\pi_i, \mu_i, \Sigma_i} \log p\left((x^1, c^1), \cdots, (x^N, c^N)\right) \\
&= \arg\max_{\pi_i, \mu_i, \Sigma_i} \sum_{n=1}^{N} \log p(x^n|c^n)p(c^n) \\
&= \arg\max_{\mu_i, \Sigma_i} \sum_{n=1}^{N} \log p(x^n|c^n) + \arg\max_{\pi_i} \sum_{n=1}^{N} \log p(c^n)
\end{aligned}$$

- The left argmax represents the mixture component modes (Gaussian)
- The right argmax represents mixing proportions.
- They can be optimized separately.

Mixing Proportions

$$p(c = i) = \frac{N_i}{N}$$

where

- N: Total number of samples
- N_i: Number of samples in cluster i

Mixture model parameters

$$\mu_i = \frac{1}{N_i} \sum_{n=1}^{N} x^n I(c^n = i)$$

$$\Sigma_i = \frac{1}{N_i} \sum_{n=1}^{N} (x^n - \mu_i)(x^n - \mu_i)^T I(c^n = i)$$

where N_i: number of samples in cluster i

- $I(c^k = i)$: Indicator function that takes value (1) if $c^k = i$, (0) otherwise.

Learning with Fully Observed Data Summary

- Data: $(x^1, c^1), \cdots, (x^N, c^N)$
- Goal: learn model parameters π_i, μ_i, Σ_i for $i = 1, \cdots, K$
- Procedure:
  - Mixing proportions: $\pi_i = \frac{N_i}{N}$, where $N_i = \sum_{n=1}^{N} I(c^n = i)$
  - Mixture model parameters:
    * $\mu_i = \frac{1}{N_i} \sum_{n=1}^{N} x^n I(c^n = i)$
    * $\frac{1}{N_i} \sum_{n=1}^{N} (x^n - \mu_i)(x^n - \mu_i)^T I(c^n = i)$

Learning on Partially Observed Data

- Given samples $x^1, \cdot, x^N$, where each $x^n = [x_1^n, \cdots, x_J^n]$ for N samples and J variables
- Assume **cluster labels** $c^1, \cdots, c^N$ are **unknown**

We have a latent variable model. (c is unobserved, only x is observed). What we know is that MLE is easy if labels were observed. And labels can be inferred, if MLE solutions are known.

EM Algorithm

Try maximizing the data log likelihood for N samples!

$$\begin{aligned}
\log \prod_{n=1}^{N} p(x^n) &= \sum_{n=1}^{N} \log p(x^n) \\
&= \sum_{n=1}^{N} \log \sum_{i=1}^{K} p(x^n, c^n = i) \\
&= \sum_{n=1}^{N} \log \sum_{i=1}^{K} p(x^n | c^n = i) p(c^n = i)
\end{aligned}$$

Difficult to optimize because c^n is missing!

EM Optimization Criterion

Maximize the 'expected' data log likelihood for N samples:

$$E_{p(c^n|x^n)}\left[\sum_{n=1}^{N}\log p(x^n, c^n)\right]$$

Iterate until convergence:

- Assign samples to clusters. (**E-step**: Infer cluster labels c^n)
- Find the cluster means. (**M-step**: Parameter estimation (π_i, μ_i, Σ_i))

(Expectation) E-Step

Assign clusters $p(c^n|x^n, \theta^{(t)})$

- Compute expected likelihood with respect to latent cluster labels

$$E_{p(c^n|x^n,\theta^{(t)})}\left[\log\prod_{n=1}^{N}p(x^n, c^n)\right]$$

- $\theta^{(t)} = \{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$ are current estimates of parameters

(Maximization) M-step

Find the parameters that maximize this quantity

$$\theta^{(t+1)} = \arg\max_{\theta} E_{p(c^n|x^n,\theta^{(t)})}\left[\log\prod_{n=1}^{N}p(x^n, c^n)\right]$$

- EM algorithm is guaranteed to converge: the log likelihood of the observed data is guaranteed to increase in each iteration.

E-step in Gaussian Mixture

- Compute cluster label probability $P(c^n = i|x^n)$ for each data point given the current parameters $\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$

$$P(c^n = i|x^n) = \frac{p(x^n|c^n = i)p(c^n = i)}{\sum_{i=1}^{K}p(x^n|c = i)p(c^n = i)}$$

M-step in Gaussian Mixture

- Compute parameters given cluster labels $P(c^n = i|x^n)$
- Mixing proportions

$$\pi_i = \frac{1}{N}\sum_{n=1}^{N}p(c^n = i|x^n)$$

  - where N is the total number of samples

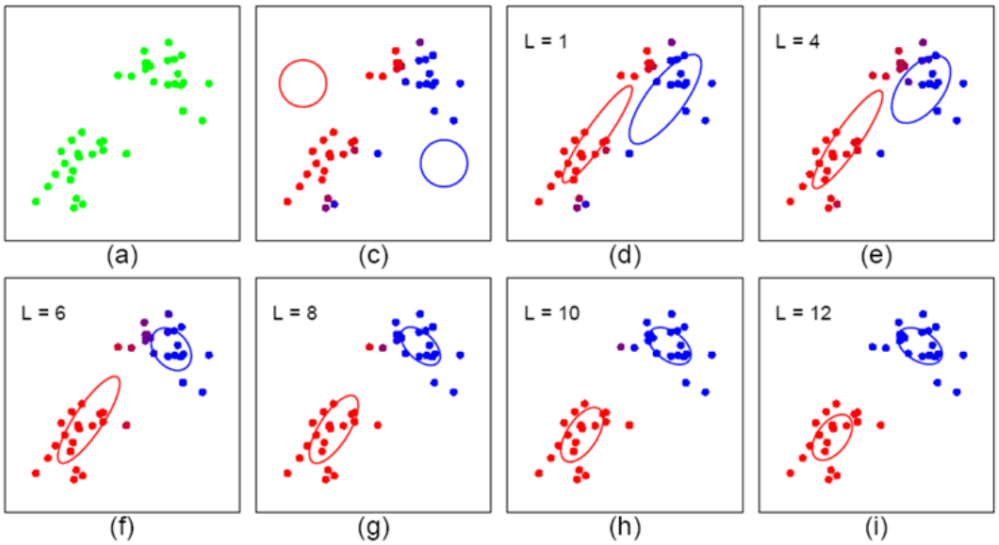
- Mixing model parameters

$$\mu_i = \frac{\sum_{n=1}^{N}x^n p(c^n = i|x^n)}{\sum_{n=1}^{N}p(c^n = i|x^n)}$$

$$\Sigma_i = \frac{\sum_{n=1}^{N}(x^n - \mu_i)(x^n - \mu_i)^T p(c^n = i|x^n)}{\sum_{n=1}^{N}p(c^n = i|x^n)}$$

Example: EM Algorithm on Gaussian Mixture

- To start: "Guess" the centroid μ_i and covariance Σ_i of each of the K clusters
- Loop:



Convergence in EM

- The log likelihood of the observed data is **guaranteed to increase** in each iteration of the EM algorithm. One can prove this monotonic increase data log likelihood.
  - Data log likelihood: $\log \prod_{n=1}^{N} p(x^n)$
  - Expected complete data log likelihood: $E[\log \prod_{n=1}^{N} p(x^n, c^n)]$
- EM converges to local minimum, and there are multiple local minimums
  - Latent variable model $\rightarrow$ objective function non-convex

Learning with Partially Observed Data Summary

- Data: $x^1, \cdots, x^N$ (note that labels c^n not observed)
- Goal: learn model parameters π_i, μ_i, Σ_i for $i = 1, \cdots, K$
- Procedure:
  - **E step:** compute $P(c^n = i|x^n)$ given current parameters $\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$
    - $P(c^n = i|x^n) = \frac{p(x^n|c^n=i)p(c^n=i)}{\sum_{i=1}^{k} p(x^n|c=i)p(c^n=i)}$
  - **M step:** compute parameters given cluster labels $P(c^n = i|x^n)$
    - $\pi_i = \frac{1}{N}\sum_{n=1}^{N} p(c^n = i|x^n)$
    - $\mu_i = \frac{\sum_{n=1}^{N} x^n p(c^n=i|x^n)}{\sum_{n=1}^{N} p(c^n=i|x^n)}$
    - $\Sigma_i = \frac{\sum_{n=1}^{N} (x^n-\mu_i)(x^n-\mu_i)^T p(c^n=i|x^n)}{\sum_{n=1}^{N} p(c^n=i|x^n)}$

EM in Practice

- Initialization of EM: set the parameters to random values
- Results can vary based on initialization
- Run EM with multiple different initializations
  - Multiple random initializations
  - Initialize with the results of simpler methods, e.g., K-means
- Select the results with the highest data log likelihood

How to Select the Number of Clsuters

- Cross validation
- Compute the likelihood of the held-out (validation) data to determine which model (number of clusters) is more accurate

$$\log p(x^1, \ldots, x^N) = \sum_{n=1}^{N} \log p(x^n|c^n)p(c^n)$$

- Nonparametric Bayesian prior
  - Place a Dirichlet process prior on mixture proportions
  - Infinite mixture models
    * Assume an infinite number of clusters in prior
    * Only a finite number of clusters are used to model data

Constructing Mixture Models

- We have focused on mixture of Gaussians
  - Clustering genes and samples based on gene expression levels
- Mixture of Bernoulli distribution for cluster discrete data
  - Cluster genome sequence data to find population structures
  - Expression quantifcation from RNA-seq reads: cluster RNA-seq reads into the ones mapped to genes
- Mixture of HMMs
  - More complex models for clustering genome sequence data

Bonus: Bernoulli Mixture Models

- Data points $x^1, \cdots, x^N \in \mathbb{R}^J$, latent clusters $c^1, \cdots, c^N$
- Data generation process:
  - $c^n = i$ with probability π_i
  - $x^n|c^n = i \sim Bern(\mu_i)$, i.e., x^n_j is a coin toss with probability μ_{ij}
- Distribution $p(x^n) = \sum_{i=1}^{K} p(x^n|c^n = i)p(c^n = i)$
- Inference $p(c^n|x^n)$

- EM algorithm
  - E step: $p(c^n|x^n;\pi_i,\mu_i)$
  - M step:

$$\pi_i^{new} = \frac{1}{N}\sum_{n=1}^{N} p(c^n = i|x^n;\pi_i,\mu_i)$$
$$\mu_i^{new} = \frac{\sum_{n=1}^{N} x^n p(c^n = i|x^n;\pi_i,\mu_i)}{\sum_{n=1}^{N} p(c^n = i|x^n;\pi_i,\mu_i)}$$

Clustering Methods: Comparison

	Hierarchical	**K-means**	**GMM**
Running Time	naively, $O(N^3)$	fastest (each iteration is linear)	fast (each iteration is linear)
Assumptions	requires a similarity / distance measure	strong assumptions	strongest assumptions
Input parameters	none	K (number of clusters)	K (number of clusters)
Clusters	subjective (only a tree is returned)	exactly K clusters	exactly K clusters

What's a Good Clustering?

- Internal criterion: A good clustering will produce high quality clusters in which:
  - the intra-class (that is, intra-cluster) similarity is high
  - the inter-class similarity is low
- External criteria for clustering quality
  - Assesses a clustering with respect to external resources
  - Gene clustering: do genes in each cluster overlap with the genes annotated similarly in Gene Ontology (GO): **GO enrichment analysis**

Summary

- Mixture models as a probabilistic version of K-means
- EM algorithm for learning mixture models